Sequences

A *sequence* is a function f defined for every nonnegative integer n, denoted by $x_n = f(n)$. Usually it is defined as an equation of the form

$$x_n = F(x_{n-1}, x_{n-2}, \ldots),$$

which is a form of *functional equation.* A functional equation of the form

$$x_n = px_{n-1} + qx_{n-2} \quad (q \neq 0)$$

is a (homogeneous) *linear difference equation of order* 2 (with constant coefficients). To solve it, try $x_n = \lambda^n$, plug into the equation, get $\lambda^n = p\lambda^{n-1} + q\lambda^{n-2}$. If $\lambda \neq 0$, get $\lambda^2 = p\lambda + q$, or $\lambda^2 - p\lambda - q = 0$. This is the characteristic equation of the difference equation. For distinct roots λ_1 and λ_2, we try $x_n = a\lambda_1^n + b\lambda_2^n$, with a and b determined by the initial values x_0 and x_1. For $\lambda_1 = \lambda_2 = \lambda$, we try $x_n = (a + bn)\lambda^n$. Higher order equations are considered likewise.

Another property we usually consider is what happen when n getting bigger and bigger, does it have a limit, i.e., does $\lim_{n\to\infty} x_n$ exist? Typically, if a sequence x_n is increasing (decreasing) but bounded above (below), then such a limit exists.
Typically we have arithmetic sequences, geometric sequences, etc. We also consider is a sequence is *periodic*, takes on (or not) certain values, general formula, other *recursive relations*, and all the like.

Typical sequences:

(i) (Fibonacci sequence) $F_0 = F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$, $n \geq 0$. Its general formula is $F_n = \dfrac{\alpha^n - \beta^n}{\sqrt{5}}$, with $\alpha = \dfrac{1+\sqrt{5}}{2}, \beta = \dfrac{1-\sqrt{5}}{2}$.

(ii) It can be shown that $(1 + \dfrac{1}{n})^n$ is strictly increasing and bounded above by 3. Its limit is the constant e.

(iii) The *arithmetic-geometric mean of Gauss*. Let $0 < a < b$. Define two sequences

a_n, b_n as follows.

$$a_0 = a, b_0 = b, \quad a_{n+1} = \sqrt{a_n b_n}, b_{n+1} = \frac{a_n + b_n}{2}.$$

It can be shown that a_n is strictly increasing, b_n strictly decreasing, and $a_n < b_n$ for all *n*. Moreover, a_n and b_n tend to a same limit.

<u>Example (Oct 2018Test)</u> Determine all sequences p_1, p_2, p_3, … of prime numbers for which there exists an integer *k* such that the recurrence relation

$$p_{n+2} = p_{n+1} + p_n + k$$

holds for all positive integers *n*.

The sequence must be a constant sequence. Using $p_{n+3} = p_{n+2} + p_{n+1} + k$ and $p_{n+2} = p_{n+1} + p_n + k,$ by subtracting, get $p_{n+3} - 2p_{n+2} + p_n = 0,$ we get the characteristic equation

$\lambda^3 - 2\lambda^2 + 1 = 0,$ with roots 1, $\frac{1 \pm \sqrt{5}}{2}$, hence

by the method of characteristic equation, the general solution to the recurrence relation is

$$p_n = A\left(\frac{1+\sqrt{5}}{2}\right)^n + B\left(\frac{1-\sqrt{5}}{2}\right)^n + C,$$

where *A*, *B*, *C* are some constants. Furthermore, as the sequence must be periodic modulo p_1 (for there are at most p^2 possibilities for (p_{n+1}, p_n) which determines p_{n+2}), infinitely many terms of the sequence are congruent to p_1 modulo p_1 (hence equal to p_1). This forces $A = 0$ (for otherwise the sequence will eventually be strictly increasing if $A > 0$ or strictly decreasing if $A < 0$) and hence $B = 0$ (for otherwise p_n cannot be an integer). This gives us the desired conclusion.

<u>Example (Chinese Mathematical Contest 1990)</u> Let $E = \{1, 2, ..., 200\}$ and $G = \{a_1, a_2, ..., a_{100}\} \subseteq E$ with the following properties:

(i) For all $1 \le i < j \le 100$, $a_i + a_j \ne 201$.

(ii) $\sum_{i=1}^{100} a_i = 10080$.

Show that the number of odd members in G is a multiple of 4, also the sum of squares of members in G is a constant.

(i) Clearly the number of odd members in G cannot be $4k+1$ or $4k+3$. Assume the number of odd members in G be $4k+2$, say $a_1, a_2, ..., a_{4k+2}$, then $a_{4k+3}, a_{4k+4}, ..., a_{100}$ are even numbers. Hence those even numbers not belonging to G are $201-a_1, 201-a_2, ..., 201-a_{4k+2}$. Now the sum of all even numbers from 1 to 200 is 10100. Thus $\sum_{j=4k+3}^{100} a_j + \sum_{j=1}^{4k+2} (201-a_j) = 10100$. Also $\sum_{j=4k+3}^{100} a_j + \sum_{j=1}^{4k+2} a_j = 10080$, solve to get $\sum_{j=1}^{4k+2} a_j = 191+402k$, contradiction.

(ii) Let $a_1, a_2, ..., a_{100}$ be members of G, then $201-a_1, 201-a_2, ..., 201-a_{100}$ are members not in G. Hence

$\sum_{k=1}^{200} k^2 = \sum_{i=1}^{100} a_i^2 + \sum_{i=1}^{100} (201-a_i)^2 = 2\sum_{i=1}^{100} a_i^2 - 402\sum_{i=1}^{100} a_i + 201^2 \times 100$, hence the value of $\sum_{i=1}^{100} a_i^2$ is fixed.

Example (Chinese Mathematical Olympiad 1988) Let $\{a_n\}$ be a sequence, with $a_1 = 1, a_2 = 2,$ and when $n \ge 1$,

$$a_{n+2} = \begin{cases} 5a_{n+1} - 3a_n & \text{if } a_n \cdot a_{n+1} \text{ even,} \\ a_{n+1} - a_n & \text{if } a_n \cdot a_{n+1} \text{ odd.} \end{cases}$$

Show that $a_n \ne 0$.

We may prove by induction $a_{2n-1} \equiv 1 \pmod 3, a_{2n} \equiv 2 \pmod 3$.

Example Let $\{a_n\}$ be a sequence, with $a_1 = -1,$ and when $n \ge 1,$ and

$$a_{n+1} = 2a_n + \sqrt{3a_n^2 + 1}, \quad n \ge 1.$$

Show that all a_n's are integers.

Try a few values and guess $a_{n+2} = 4a_{n+1} - a_n$ and prove it by induction. Indeed we have $a_n^2 - 4a_n a_{n+1} + (a_{n+1}^2 - 1) = 0$ and $a_{n+2}^2 - 4a_{n+2}a_{n+1} + (a_{n+1}^2 - 1) = 0,$ thus a_n, a_{n+2} are (distinct) roots of the equation $x^2 - 4a_{n+1}x + (a_{n+1}^2 - 1) = 0.$ From Vieta, get

$a_{n+2} + a_n = 4a_{n+1}$ for all n.

Example Let $\{a_n\}, \{b_n\}$ be sequences with the following properties:

$$a_0 = 1, a_1 = 1, a_{n+1} = a_n + 2a_{n-1}, \quad n \geq 1,$$
$$b_0 = 1, b_2 = 7, b_{n+1} = 2b_n + 3b_{n-1}, \quad n \geq 1.$$

Show that the two sequences have no common term except the "1".

Try modulo 8, get 1, 1, 3, 5, 3, 5, 3, 5,… for a_n and 1, 7, 1, 7, 1, 7,… for b_n. Hence conjecture $a_{2n+1} \equiv 3 \pmod 8, a_{2n+2} \equiv 5 \pmod 8$ and $b_{2n+1} \equiv 1 \pmod 8, a_{2n+2} \equiv 7 \pmod 8$.

Example (Putnam 1963) Let $f(n)$ be a strictly increasing sequence with $f(2) = 2$ and when m, n relatively prime, $f(mn) = f(m)f(n)$. Show that $f(n) = n$ for all n.

First $f(1) = 1$, be taking $m = n = 1$. Now $f(3) > 2$, take $m = 3, n = 7$, get $f(3)f(7) = f(21) < f(22) = f(2)f(11) = 2f(11) < 2f(14) = 2f(2)f(7) = 4f(7)$, get $f(3) < 4$, thus $f(3) = 3$. Assume now n_0 is the smallest number such that $f(n_0) > n_0$, that means $n_0 \geq 4$, with $f(1) = 1, f(2) = 2, ..., f(n_0 - 1) = n_0 - 1$, but $f(n_0) > n_0$. This will imply $f(n) > n$ for $n \geq n_0$. Now if n_0 is odd, then 2 and $n_0 - 2$ relatively prime, we have $f(2(n_0 - 2)) = f(2)f(n_0 - 2) = 2(n_0 - 2)$, but we should have $f(2(n_0 - 2)) > 2(n_0 - 2)$. If n_0 is even, consider 2 and $n_0 - 1$, get $f(2(n_0 - 1)) = f(2)f(n_0 - 1) = 2(n_0 - 1)$, again the same conclusion!

Example Show that the term $a_n = \sum_{k=0}^{n} \binom{2n+1}{2k+1} 2^{3k}, n \geq 0,$ is never divisible by 5.

Check that $a_0 = 1, a_1 = 11$, and

$$a_n = \frac{(2\sqrt{2}+1)^{2n+1} + (2\sqrt{2}-1)^{2n+1}}{4\sqrt{2}}$$
$$= \frac{(2\sqrt{2}+1)(9+4\sqrt{2})^n + (2\sqrt{2}-1)(9-4\sqrt{2})^n}{4\sqrt{2}},$$

with $(9+4\sqrt{2})+(9-4\sqrt{2})=18$ and $(9+4\sqrt{2})(9-4\sqrt{2})=49$.

We thus have a recursive relation

$a_{n+1}=18a_n-49a_{n-1}\equiv 3a_n+a_{n-1} \pmod 5$.

Checking the remainder mod 5, we have 1, 1, 4, 3, 3, 2, 4, 4, 1, 2, 2, 3, 1, 1, we thus have a periodic sequence $a_{n+12}\equiv a_n \pmod 5$.

Remark. Indeed for a linear recursive sequence $a_{n+k}=c_1a_{n+k-1}+c_2a_{n+k-2}+...+c_na_n$, it is periodic modulo any natural number *m*. For we consider the sequence of length *k*, $(a'_n, a'_{n+1}, ..., a'_{n+k-1})$, with $a'_i \equiv a_i \pmod m$, there are m^k of them, thus among m^k+1 sequences, two of them are equal,

$(a'_{n_0}, a'_{n_0+1}, ..., a'_{n_0+k-1})=(a'_{n_0+s}, a'_{n_0+s+1}, ..., a'_{n_0+s+k-1})$, or

$a'_{n_0+k}=a'_{n_0+k+s}, a'_{n_0+k+1}=a'_{n_0+k+s+1}, ...,$ thus $\{a'_n\}$ is periodic of period s.

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